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ORIGINAL ARTICLE

Linking the Kansei Food Model to the General Affective Engineering Model

– An Application on Chocolate Toffee Fillings –

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Abstract: The methodology of Kansei Engineering can grasp consumer's subjective affective impressions about a product and turn it into concrete product solutions. The Kansei Food model is a specialized model doing this for food products. The aim of this paper is to interlink it with the more general model on Affective Engineering as well as to validate the findings by applying them in a case study. The general Kansei Engineering model and the Kansei Food model were analyzed, and the key parts of both models merged to a hybrid model. This model is then applied in a study on a development project for chocolate toffee fillings for sport applications. The case yielded valid results and gave an input to a parallel food development process. In conclusion, the Kansei Food model fits together with the general Kansei model. Hence, standardization makes it possible to have a more detailed look on sequential steps. Also, it becomes possible to transfer tools from other branch models (e.g. from automotive industry) in food industry and vice versa.

Keywords: *Product development, Affective design, Chocolate, Toffee fillings, Food design*

1. INTRODUCTION

Each industrial branch has its own product development methods. Even if they have certain similarities, they are also dedicated to their purpose. Food design is distinct from packaging design and both are distinctly different from the design of durable consumer goods. Yet, within each domain there is of course also variety. Within each, dedicated approaches have been developed to apply affective product development methods, e.g. Kansei Engineering. Application of Kansei Engineering in food has been somewhat limited, while for durables it is more commonplace.

Typically, development methods in the food industry mean that a food designer (or a team of food designers) prepare a number of prototypes and test them with potential consumers [1]. Several feedback loops allow the designer to improve the product until it is satisfactory [1].

One weakness in this process is that many of the decisions are made consciously or unconsciously by the designer and only the final result is evaluated by customers [2]. Most of those decisions are never documented. Even if the design team is experienced, they still are biased by their own background as well as by corporate culture [3].

Consequently, not all possible concept solutions reach prototyping stage. Most of them are filtered out by the designer, based on availability of resources, company image, cultural background, etc. In the end only a small selection of possible prototypes is presented to a test panel. With an experienced design team and panel this methodology still might not have a noticeable effect on the results, but smaller companies with less experienced designers and taste panels may struggle to find a good product composition.

From a scientific experimenter's point of view these samples are not representing the full range of the solution space. Significant product compositions could be missed, and optimal products might remain undiscovered. To avoid this, a methodology would be desirable for choosing prototypes in a way that represent the whole solution space. This would enable the designer to optimize the products in accordance with consumers' taste preferences.

There is however another aspect in this context. In addition to good quality [4], today's food products must also express certain affective values and hedonic aspects (such as luxuriousness and healthiness) [5]. To be desired (and obtained), products need to possess the right affective properties [6]. The consumer goods industry has understood this for about 20 years and utilizes today methods to cope with those affective aspects in product development [7]. Hence, integration of those methods in the food industry could complement traditional methods and lead to superior products.

In the 1970 Japanese researchers were forerunners for the then new product development method of Kansei Engineering. Kansei means roughly translated "feeling" or "emotion" and the methodologies developed then aimed to create links between consumers subjective (affective)

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impressions and concrete product solutions [8]. Those methods require systematically testing suitable combinations of product properties against user affective needs and finding an optimum. Today these methods are mostly used in the automotive industry, electronic industry and consumer goods development. Even in food industry several applications can be found [9].

This paper is aiming towards interlinking the highly specialized Kansei Food Model and the more general model on Affective Engineering thus enabling the Kansei Food model to be set into a relationship to other Kansei models used in, for example, the automotive industries. Also, the findings are proven and validated by applying them in a case study.

2. THEORETICAL FRAMEWORK

2.1 The general Kansei Engineering Model

Many industries conducting product development need to consider consumers' affective needs. In the 1970s a new area of research was founded in Japan; later coined as Kansei or Affective Engineering [10, 8]. Methods within this research area aim towards integration of affective values in products.

In 2004, Schütte, Eklund, Axelsson and Nagamachi proposed a general model on Kansei Engineering procedure. It includes a list of methods and tools that can be used in industry's existing product development processes. The model is a summary of all current Kansei Engineering types. It is presented in Figure 1.

Unlike other customer insight techniques, the Kansei Engineering procedure starts with the definition of a consumer domain by defining, for example, typical consumer personas [11] paired with products they use in all aspects of their daily life (from toothpaste to the airline

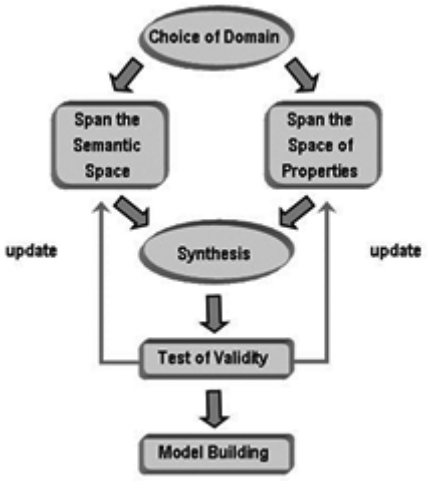


Figure 1: The Kansei Model [13]

they are flying). Hence, it is important to understand in depth the consumer's values and to a certain degree being able to predict their consume behaviour.

Based on that domain the idea behind the product can be described from two different perspectives. The Semantic Space collects consumers' affective needs of the new product. The description of product properties in turn includes the physical properties of the product which can be manipulated by the product designer.

The synthesis phase is unique to the Kansei Engineering method. The semantic and physical properties are linked to each other using mathematical statistical tools [12-14]. This makes it possible to determine which of the product properties evokes which affective impact. Product designers can then determine which product features to include and in what way to accomplish the intended affective impression.

After these steps have been carried out, it is possible to conduct a validity test, including several types of post-hoc analyses. A prototype is tested against a group of customers. As a result of this step, the Semantic and physical Properties Spaces may be updated, and the synthesis step may be rerun. This process is continued until the methods yield satisfying results.

In the final step a mathematical model is built describing the link between the Semantic Space and the Space of Properties. This model is then used for the development of a product specification list which in turn is fed back to the specific product development process in the company [14].

This method has been successfully applied in different industrial branches for very different products. This demonstrates the versatility of the model. The following examples illustrate the generalisability of the method.

The vehicle industry has been leading in the integration of affective needs. One of the earliest examples of the application of Kansei Engineering is the Japanese Manufacturer Mazda, which developed its 1986 model MX5 (Miata) using the Kansei Engineering Method [15]. European vehicle companies have also applied the methodology; the Swedish-Japanese warehouse truck manufacturer BT-Toyota used it to develop a good "driver feeling" and improved "quality feeling" of their products [16]. Other manufacturers evaluated interior components such as rocker switches [17, 18], and rubber keys [19]. In the electronics industry, the applications were foremost in improving user experience such as sound design [20], image quality [21-23], or customer acceptance (smart watches for the elderly) [24]. Other examples come from the clothing industry (bras) [15], shoes [25] and fashion. There are also a number of studies using Kansei Engineering in the food industry

(see following section), but the product development methods used there are different from those used by the consumer goods industry for hardware design.

2.2 The Kansei Food Model

It is clear from the standpoint of product design that not only functionality and usability has to be taken into account, but also the emotional values placed on a product [26]. This fact is even more apparent when the product is food, because eating food involves all five senses in a very direct manner [6]. Based on the idea behind Kansei Engineering, Ikeda, Nagai & Sagara [27] propose a Kansei Food model that allows linking the food product's characteristics with the consumer's sensorial perceptions. The model has later been extended to incorporate the impact of personal attributes [28]. Figure 2 shows the Kansei Food model.

The model has two distinct parts. On the left part, attributes related with the food are considered. On the right part, sensorial descriptors, related to perceptions of individuals when eating the food, are listed. The model can be followed using two paths: the upper path considers the intrinsic attributes of food, whereas the lower path considers the extrinsic attributes.

Intrinsic attributes can be proportion of ingredients – such as percentage of palm oil, or percentage of salt in the recipe [29] – or measurements done with some device or technique – such as odor intensity based on gas chromatography [30], or concentrations of volatile and non-volatile components [27]. Although the act of eating involves all five senses, intrinsic attributes are more connected with taste, smell and touch.

Extrinsic attributes, on the contrary, have no direct relationship with the food, but appear together with the food, thus also influencing the perception of the food by individuals. Extrinsic attributes can be, for instance, related with the packaging or physical elements attached to the food [31]. This is why extrinsic attributes are more connected with eyesight than intrinsic attributes.

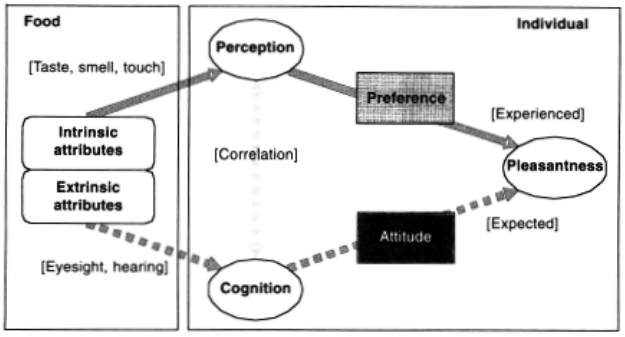


Figure 2: The (modified) Kansei Food Model [27] based on Ito's model [39]

The right part of the model incorporates the sensorial descriptors conveyed by the product. When using intrinsic attributes (upper part of the model), the sensations are more linked with the experience we have when eating the product, whereas when using extrinsic attributes (lower part of the model), the sensations are more related with our expectations regarding the product. Obviously, both paths (upper and lower, experienced and expected) are correlated one with the other, creating an overall impression.

Whereas food attributes (either intrinsic or extrinsic) are usually collected based on technical characteristics of the food, the sensorial descriptors are taken from a long list based on, for example, appearance, touch, odor, flavor, and texture. This initial long list is normally reduced to a more manageable one (having something from ten to twenty descriptors). The reduced list is often also summarized in by a small number of items, being the first components from a principal component analysis (PCA).

3. THE HYBRID KANSEI ENGINEERING FOOD MODEL – A COMBINATION OF THE KANSEI ENGINEERING MODEL WITH THE KANSEI FOOD MODEL

When comparing the Kansei Engineering model (Figure 1) and the Kansei Food model (Figure 2) it becomes obvious that they have a lot in common. Both create mathematical relationships between intangible affective values in the user's mind and physical properties of the product. While the Kansei Food model does it in a highly specific way for food products, the Kansei Engineering model is a universal model. By mapping the Kansei Food model to the Kansei Engineering model, it becomes possible to identify separate steps and by this standardize the procedure. Each of the steps is associated with tools from the Kansei Food Model. This makes it possible to test each tool and optimize it separately from the process or replace it with alternative, more suitable tools.

A proposed visualization of the Kansei Engineering Hybrid model is presented in Figure 3. It has two main areas: the identification of attributes (either intrinsic or extrinsic), which is equivalent to the establishment of the Space of Properties; and the collection of sensory descriptors, equivalent to the development of the Semantic Space. Several techniques from both parent models can be used to describe the arosen spaces. Both areas are interlinked in a synthesis step. A possible model is found, validated and finally presented as the main finding.

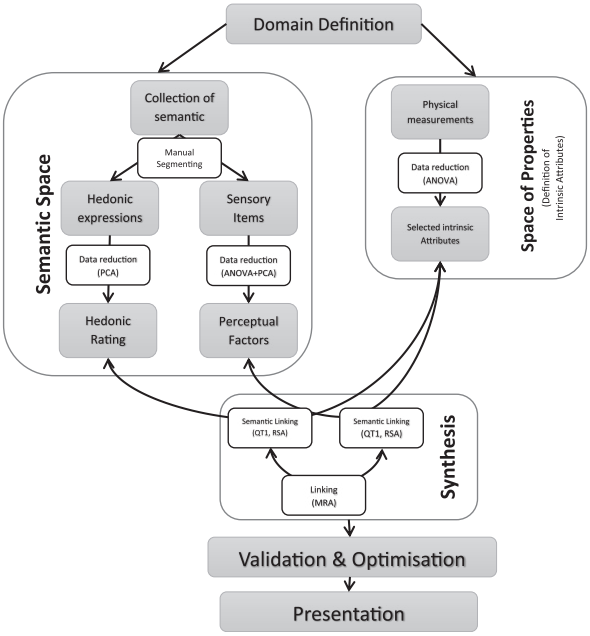


Figure 3: Hybrid Kansei Engineering Food Model

To establish the Semantic Space, affective descriptions of products are collected. The validity of tools within this method is confirmed by Osgood, Suci, & Tannenbaum [32]. While hardware products usually require the evaluation of hedonic values, food has an additional dimension: sensory items. Hence the Semantic Space was enhanced using a sensory dimension from the Kansei Food model. These are mostly verbal terms, sematic expressions and/or sensory metaphors. A manual segmentation step separates sensory items (upper path of the Kansei Food Model), which are closely related to the physical impression of the product such as taste and mastication, and the hedonic expressions (lower path of the Kansei Food Model), which are related to subjective preferences and personal values. Examples of sensory items are: airy, creamy, smooth, while typical hedonic expressions would be: delightful, like it and healthy. By using data reduction tools both of those sets will turn into a short list with expressions possessing the highest affective impact. Several methods are available for this. According to Ueda et al. [33] Principal Component Analysis can be used. Experience from other Kansei Engineering studies show that affinity diagram is a simpler and quicker yet less precise tool [34].

In parallel, physical measurements affecting the subjective experience of the food products are collected. Similarly, as described in the Semantic Space, the most influential are condensed into a short list. In this case principal component analysis is also used in the Kansei Food model. In the Kansei Engineering context Pareto charts can be used [34] whereas the Kansei Food model uses manual expert selection tools.

Subsequently both sections of the Semantic Space are linked to the Space of Properties. In Ikeda's Kansei Engineering Food model this was done using a complex system of different tools such as multiple regression analysis and principal component analysis. However, other methods have been used in consumer industry such as quantification theory type 1 (QT1) [35], rough sets analysis [12], ordinal linear regression analysis and mixed effects ordinal linear regression analysis [14] and others. It is obvious that this is a complex task requiring expertise and time resources. KESo software [36] can automate significant parts of the customer data collection and evaluation using the latter named evaluation methods. Hence, KESo software was used for the case study presented in the following section.

4. AFFECTIVE DESIGN OF CONFECTIONARY PRODUCTS

This case study was conducted in cooperation with one of the leading confectionary manufacturers in northern Europe. The confectionary market is dominated by big international concerns featuring known brands. The company prevails in this competition by possessing a number of well-known local brands and its innovativeness. There are constant activities to improve their products and introduce new taste experiences in their already established brands. One central concept is to test new fillings in limited ranges of their products.

The aim of this case was to determine the affective impact of new innovative ingredients in the filling of toffees. The filling is intended to be used in a chocolate snack consumed in a sports context.

4.1 Structure of the study

The study was carried out in accordance to the modified Kansei Engineering Food model as presented in Figure 3. In order to improve readability, a few details about the Kansei Engineering procedure were omitted, but can be found in [13] or [14]. Firstly, a domain was chosen. From this the Semantic Space and the Space of Properties were determined.

To establish the Semantic Space, emotional words were collected. These words derived from previous studies [37, 38], advertisements for competing products consumed at sport occasions and a participating company's own development team. In order to complement these data a pilot study was performed, asking 25 consumers to semantically describe their impression when eating the specific products of this product range. These words were then screened manually and separated into hedonic expressions

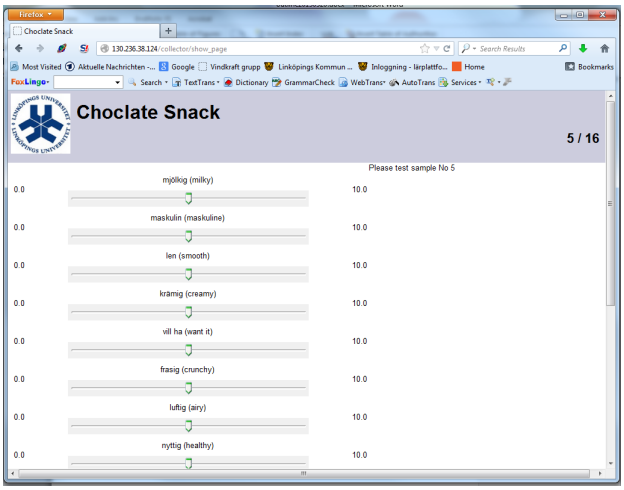


Figure 4: Data collection using KESo software

and sensory items. After applying the affinity diagram technique the number of semantic terms was reduced and expressions with overlapping meanings eliminated.

In parallel the product development division of the participating confectionary manufacturer identified key ingredients and combinations thereof with high emotional impact. In cooperation with the researchers an experimental plan was established and the selected prototypes produced.

Kansei Engineering Software (KESo) [36] was used for establishing a data collection homepage. The design of the experimentation followed the Code of Ethics of the World Medical Association. At front, the participant's informed consent was obtained for the following experimentation. Further, participants were asked to provide personal information such as: age, gender and frequency of chocolate consumption. In the actual data collection, the consumers were provided an assortment box containing samples of individually numbered chocolates containing different samples.

Despite the different flavors the samples looked identical. Further on this site they were asked to take a specified sample and test it. The participants were asked to put the sample in their mouth and crack it open by biting on it, and leave the filling on the tongue for a short while before evaluating their personal impressions on visual analogue scales in a position which "feels" right. KESo provided instant results by applying QT1 analysis [35]. Figure 4 shows the data collection software and a typical collection situation.

4.2 Study Results

As a result of the semantic data collection a total number of 151 expressions were collected. Of those 55 expressions were hedonic expressions, 32 were sensory items (20 described the texture and 12 the taste). The rest were either variants of expressions (such as sweet, sugary, sugar) or deemed without interest. After the affinity analysis the following words remained and were used in the subsequent data collection (Table 1).

Potential intrinsic attributes were collected and selected. Eventually, different recipes of combinations of ingredients were prepared. The ingredients chosen were caffeine, three different flavours, yoghurt and chilly spice to produce a hot feeling. To get best results a reduced factor table that assured orthogonality among properties was developed. Table 2 shows the 12 combinations which were tested by the participants.

In the synthesis phase, a Kolmogorov-Smirnov test ensured the data was normally distributed. Also, it was checked whether the gender, the age and the nationality (Swedish and Spanish) made a difference in how the test samples were perceived. As for gender significant differences were found only for the word "smooth" in sample 1. The age range of the participants was quite narrow. Hence, no significant difference could be observed. For the nationality, the Levene's test showed three significant differences regarding the variances. However, no significant differences were found for the means of the nationality rankings in a t-test. As a consequence, all data was pooled for further evaluation.

Table 1: Affective terms used

Hedonic expressions		Want to have
		Healthy
		Energy
		Masculine
		Like it
Sensory items	Texture	Smooth
		Creamy
		Crunchy
		Airy
	Taste	Milky
		Nut-like
		Refreshing

Table 2: Selection of stimuli for the Chocolate Case

Sample No.	Caffeine	Flavour	Yoghurt	Spice
1	Yes	Salt	No	Chilli
2	Yes	Salt	No	None
3	No	Salt	Yes	Chilli
4	No	Salt	Yes	None
5	No	Nut	No	Chilli
6	No	Nut	No	None
7	No	Nut	Yes	Chilli
8	No	Nut	Yes	None
9	Yes	Fruit	No	Chilli
10	Yes	Fruit	No	None
11	No	Fruit	Yes	Chilli
12	No	Fruit	Yes	None

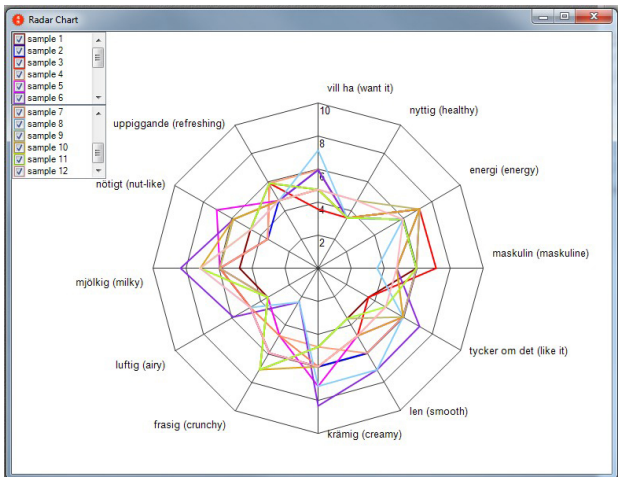


Figure 5: Emotional profiles for each product sample

Figure 5 shows the average rating values plotted in a radar chart. These presentations are commonly known as emotional profiles of the samples [13]. The emotional profiles vary widely between the samples as a result from the long ranging spanning of the Space of Properties.

4.3 Principle Component Analysis

A principal component analysis was conducted on the semantic expressions. Because of the different nature of sensory items and hedonic expressions two separate evaluations were done. Figure 6 shows the results. The upper loading plot displays the hedonic expressions. Except for "want it" and "like it" they represent own dimensions.

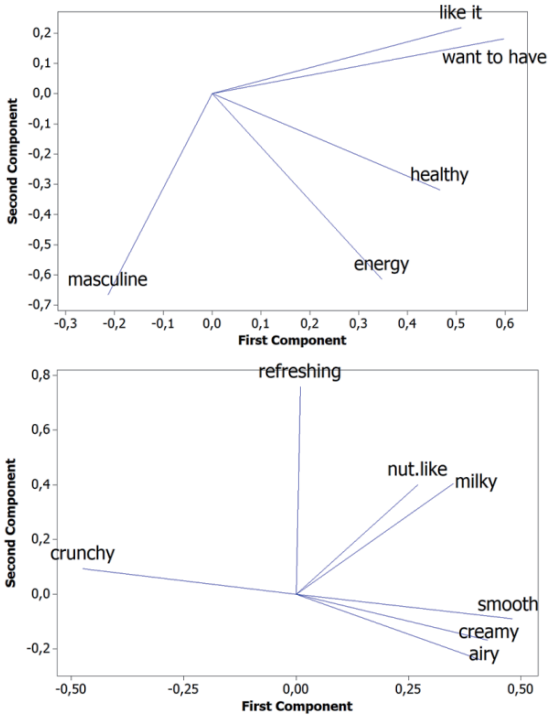


Figure 6: PCA results for sensory items (above) and hedonic expressions (below)

Table 3: Assembly of all QT1 results

	Kansei word	Caffeine		Taste			Yoghurt		Spice	
		Yes	No	Salt	Nut	Fruit	Yes	No	No	Chilli
Hedonic expressions	Want it	+	-	-	+	-	+	-	+	-
	Like it	-	+	-	+	+	-	+	+	-
	Healthy	+	-	+	+	-	+	-	+	-
	Energy	-	+	-	+	-	-	+	-	+
	Masculine			+	-		-	+	-	+
Textures	Smooth	+	-	-	+	-	+	-	+	-
	Creamy				+	+	-	+	+	-
	Crunchy	+	-	+	-	+	+	-	-	+
	Airy			-	+	-	+	-	+	-
	Milky	-	+	-	+	-			+	-
	Nut-like	-	+	-	+	+	-	+		
	Refreshing			+	-	-	+	-	-	+

"+"/"-" means positive or negative impact of the respective category on the average rating. Empty boxes mean that the property is not significant.

The bottom loading plot displays the sensory items. Creamy, smooth and airy appear close in the Semantic Space; and these three sensory items are perceived as contrary to crunchy. Nut-like and milky are also close together. Refreshing stands quite alone in the Semantic Space, and this is the only word with a different location depending on the country.

4.4 Quantification Theory Type 1

The evaluation using a special form of regression analysis, quantification theory type 1 (QT1) [35] is a quite common tool in Kansei Engineering data evaluation [9]. Table 3 shows a condensed form of all the QT1 results in one table. Although there were few minor differences in the ratings of the Spanish and Swedish groups, it was possible to find a common pattern for both groups. This implies that a chocolate product designed for either market will be perceived similarly on the other market as well. The results in Table 3 were derived from the pooled data from both national test groups. For increased comprehensiveness and due to NDA regulations with the company the exact category scores retrieved from QT1 were replaced with general tendencies. This does not affect the interpretability of the results in this context.

4.5 Conclusions of the Case Study

Applying new hybrid Kansei Engineering Food model (Figure 3) yielded relevant results. The summary given in Table 3 facilitates the design of products based on the desired conveyed characteristics. For instance, a filling that is both desired (want it) and perceived as healthy should have flavours of caffeine, and nuts, possess a yoghurt note and contain no chili. The results can also be used to determine which ingredients or combinations to avoid as well as see how a certain ingredient is correlated to a subjective impression. It can also be used to emphasize one or several impressions. For example, if the impression of creamy and milky should be achieved,

the taste needs to be nut-like and without chili. If the target group for the product are men, masculinity should be boosted, and chili and salt preferred ingredients. Those ingredients should be avoided if female buyers are targeted.

5. DISCUSSION

Kansei Engineering is relatively new and not only a toolbox, but also part of the Japanese product development philosophy. Its reductionist approach enables designers to structure and communicate their work and introduce and test new tools for each step. Food design on the other hand has mostly been carried out using holistic approaches. The arising question is: Can food experience be treated using reductionist models? And if "yes", what role can Kansei Engineering play in conventional food development processes?

In fact, many engineering methodologies are based on reductionist models (often mathematical) which are representing more complex real artefacts. Thus, because real world structures are excessively complex to include all its properties in detail, those models account for the most influential factors and neglect factors of subordinate nature making engineering design swift, accurate and documented. Even food design can participate from these advantages using Kansei Engineering, but the creative design process cannot be replaced. Instead it can be used as a design support and validation tool. In the food design process it can be used at the start for benchmarking and setting the affective specifications, during the detailed design stage it can help to determine the whole design space and in the end it can be used to verify whether the targeted specifications were met by the new product concept and if not, why.

Another question arising during the merging of the two models was if the result should serve as a new model. In fact the authors chose to call it a "hybrid model". This model however is not detailed enough to carry out complex Kansei evaluations on food. Comparing the hybrid model (Figure 3) with the Kansei Food model (Figure 2) shows a difference in complexity. Hence, the hybrid model should be seen as evidence that the general Kansei Engineering procedure and the Kansei Food model are comparable models. It makes it possible to find new tools for individual steps and to transfer tools from other models (e.g. from the automotive industry) in the food industry and vice versa.

The functionality of the hybrid model is illustrated on an example study using confectionary products. Since this was a real study carried out for a leading Scandinavian producer, the data was anonymized based on the companies' requirements. Still the data used are original and allow sufficient interpretation to confirm the functionality of the Kansei hybrid model.

6. CONCLUSION

The Kansei Food Model is a validated model which has successfully been used in the Japanese food industry. The Kansei Engineering model is a result from an extensive literature study and summarizes the procedure of Kansei Engineering investigations mostly on industrially produced consumer products. Once the key parts of each model were identified, it became possible to reassemble them in a way visualizing the working procedure. This shows that both models are practically describing identical working procedures.

The purpose of this research was not to replace any of the models, but to show that they are related and can be interlinked. Hence, the question arising was how it would be possible to interlink those models. After enhancing the Semantic Space in the general Kansei Engineering model, both models fit together. A new hybrid model was produced. This new model sets the Kansei Food model in a bigger context among other procedures and methodologies within Kansei Engineering. It also makes the procedure more comprehensive since it is comparable with other Kansei models from other areas such as the automotive and electronics industries.

Also, this standardization makes it possible to have a more detailed look on sequential steps of the model. Now, new tools and methodologies for those steps can be either newly developed or used from other specialized models (e.g. automotive industry). Also testing new tools keeping the rest of the model stable makes it easier to improve further developments.

Applying the newly assembled model on toffee fillings shows that the results are comparable with results from the Kansei Food model. Hence, the researchers draw the conclusion that the key factors of both models were correctly identified and assembled.

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